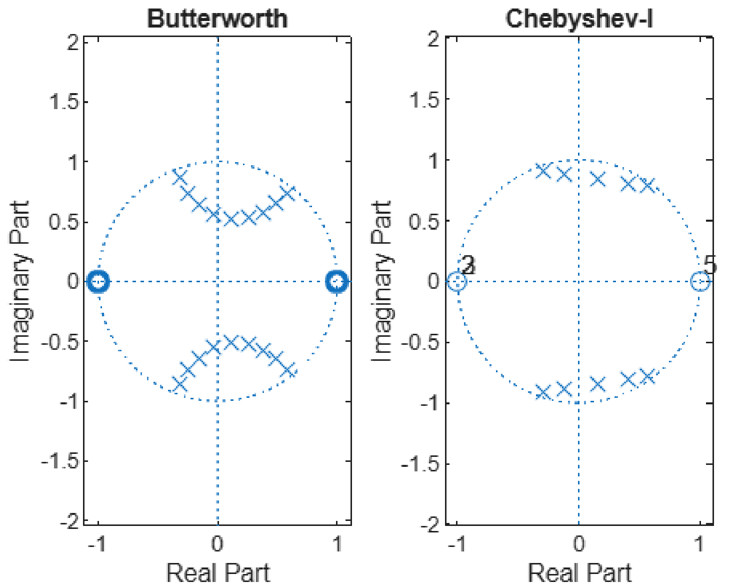
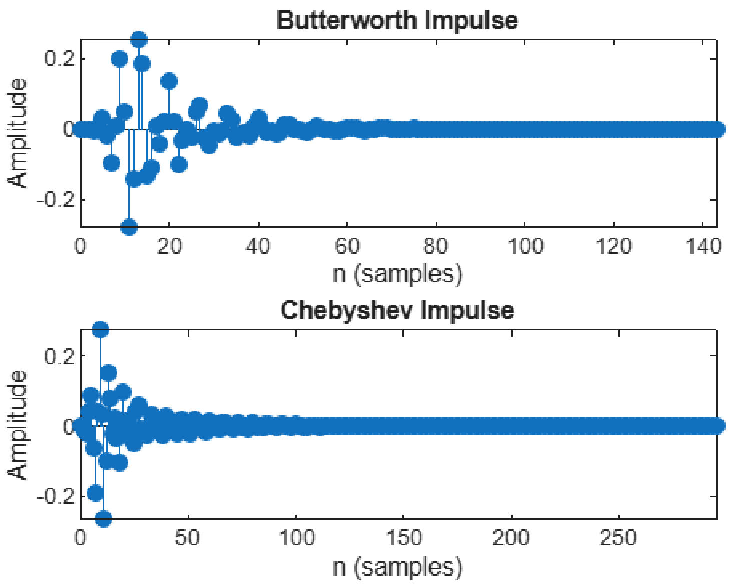
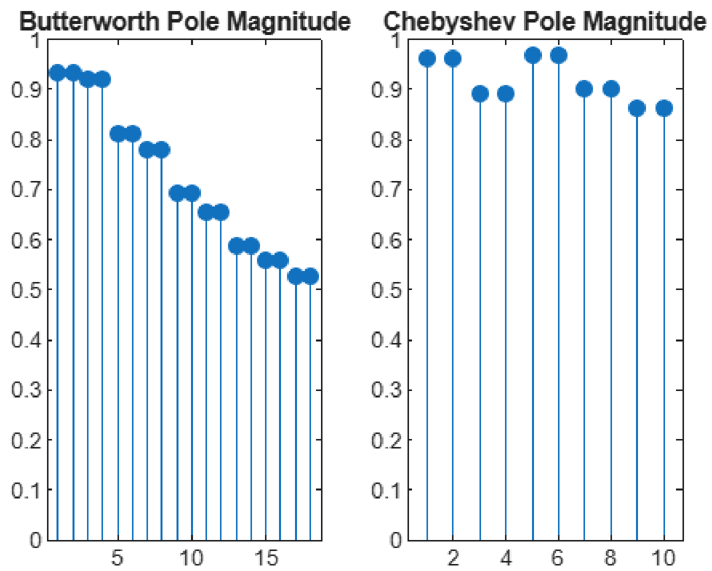
EX NO:	Performance Analysis of IIR Filters
DATE:	3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB.

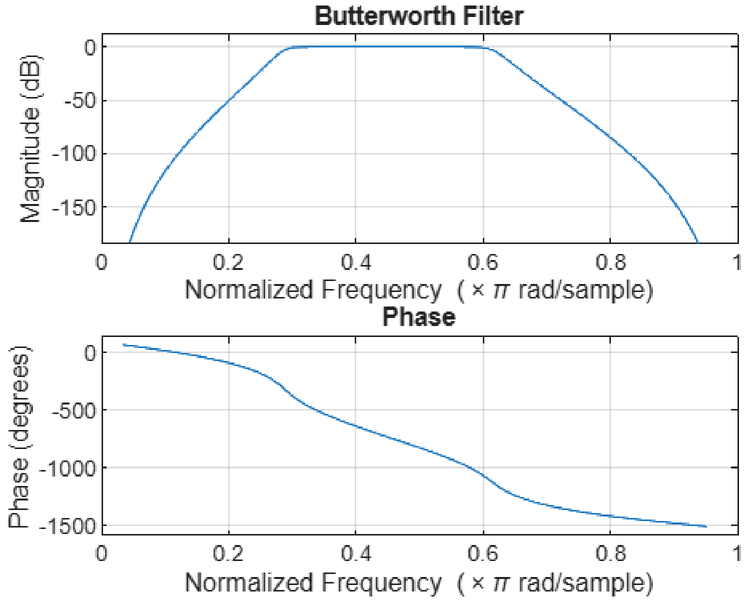
```
clc; clear; close all;

% Specifications

Fs=2000; Rp=1; Rs=40;

Wp=[300 600]/(Fs/2);

Ws=[200 700]/(Fs/2);


% Butterworth Filter

[n1,W1]=buttord(Wp,Ws,Rp,Rs);

[b1,a1]=butter(n1,W1,'bandpass');


% Chebyshev Type-I Filter

[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);

[b2,a2]=cheby1(n2,Rp,W2,'bandpass');


% Magnitude Response

figure;

freqz(b1,a1); title('Butterworth Filter');


figure;

freqz(b2,a2); title('Chebyshev Type-I Filter');


% Group Delay

figure;

grpdelay(b1,a1); hold on;
```

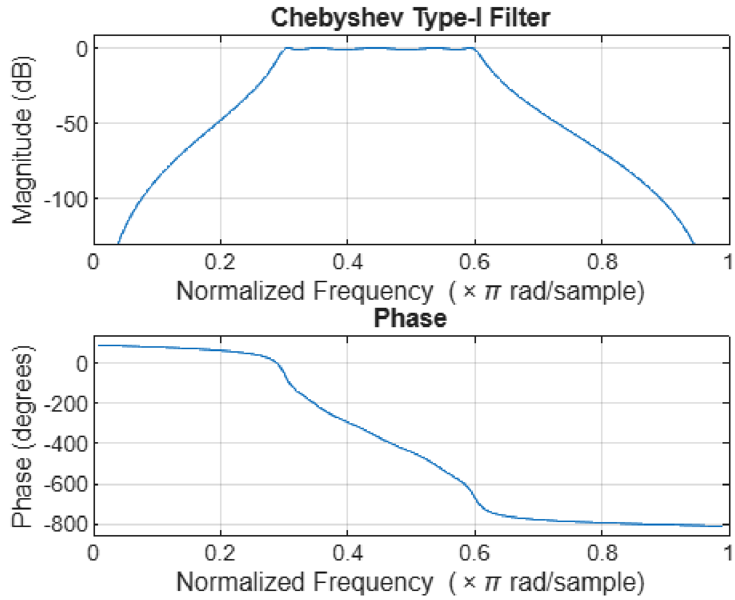
```
grpdelay(b2,a2);
legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')

% Impulse Response
figure;
subplot(211),impz(b1,a1),title('Butterworth Impulse')
subplot(212),impz(b2,a2),title('Chebyshev Impulse')

% Pole Magnitude
figure;
subplot(121),stem(abs(roots(a1)),'filled')
title('Butterworth Pole Magnitude')

subplot(122),stem(abs(roots(a2)),'filled')
title('Chebyshev Pole Magnitude')

disp(['Butterworth Order = ',num2str(n1)])
disp(['Chebyshev Order = ',num2str(n2)])
```

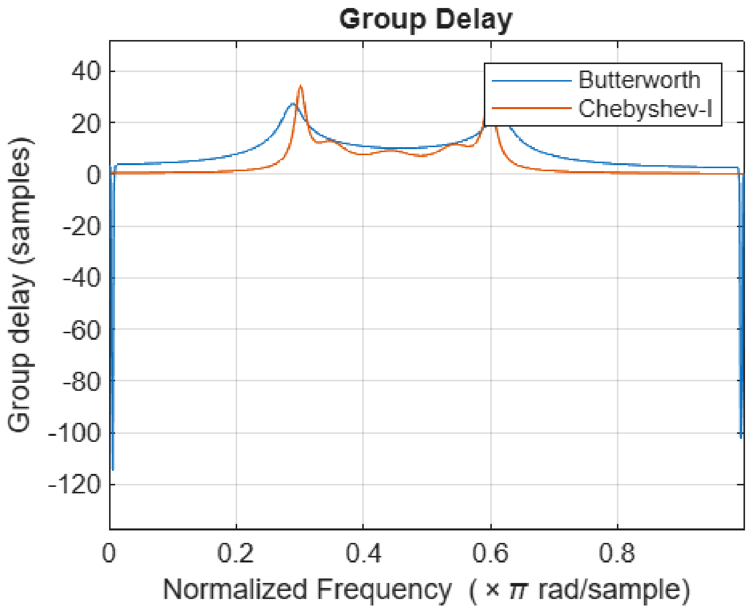

Butterworth
Chebyshev-I
Imaginary Part
Real Part
3
5


Butterworth Impulse
Amplitude
n (samples)
Chebyshev Impulse
Amplitude
n (samples)


Butterworth Pole Magnitude
Chebyshev Pole Magnitude


Butterworth Filter
Magnitude (dB)
0
-50
-100
-150
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-500
-1000
-1500
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Chebyshev Type-I Filter
Magnitude (dB)
0
-50
-100
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-200
-400
-600
-800
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)


Group Delay
Butterworth
Chebyshev-I
Group delay (samples)
40
20
0
-20
-40
-60
-80
-100
-120
0
0.2
0.4
0.6
0.8
Normalized Frequency (× π rad/sample)